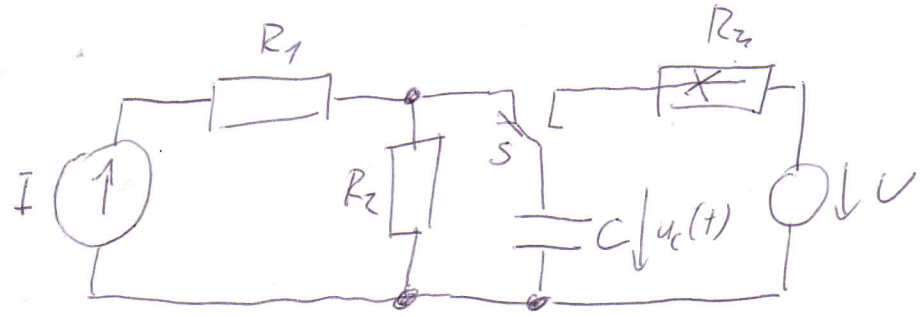


$R_1 = 1\,k\Omega$

$R_2 = 2\,k\Omega$

$R_3 = 1\,k\Omega$

$C = 1\,\mu F$

$I = 1\,mA$

$U = 1V$

$$C\frac{du_c}{dt} + \frac{u_c - U}{R_3} = 0$$

$$C\frac{du_c}{dt} + \frac{u_c}{R_3} = \frac{U}{R_3}$$

$$R_3 C\frac{du_c}{dt} + u_c = U$$

$$R_3 C\frac{du_c}{dt} + u_c = U$$

$$R_3 C \lambda + 1 = 0$$

$$\lambda = \frac{-1}{R_3 C}$$

$$u_0 = k e^{\lambda t} = k e^{-\frac{t}{R_3 C}} = k e^{-\frac{t}{\tau}} \; ; \; \tau = R_3 C$$

$$u_c(t) = u_0(t) + u_p(t)$$

$$u_p(t) = u_0(\infty) = U$$

$$u_c(t) = k e^{\frac{-t}{\tau}} + U$$

$$u_c(0) = k + U = U_1$$

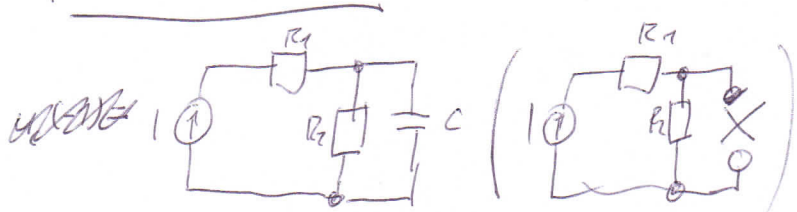
$$K = U_1 - U$$

$$u_c(t) = (U_1 - U) e^{\frac{-t}{\tau}} + U \; ; \; \tau = R_3 C$$

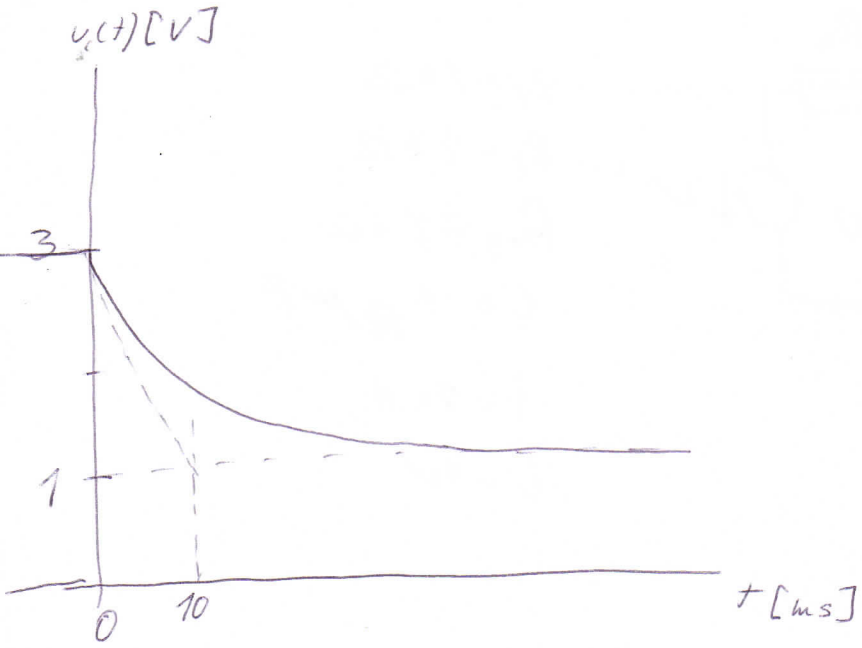
$$u_c(t) = 2V e^{\frac{-t}{1k\Omega \cdot 1\mu F}} + 1V$$

$$u_c(t) = 2 e^{\frac{-t}{0{,}001}} + 1$$

před sepnutím



$$U_1 = I \cdot (R_1 + R_2) = 1mA \cdot (1k\Omega + 2k\Omega) = 3V$$


$v_c(t)$ [V]
3
1
0
10
t [ms]